Relations as a module over the target prime field

Project to the cyclic order-n component before interpreting factor-base entries as logarithms.

$$G_n = \langle P \rangle \cong \mathbb{Z}/n\mathbb{Z}, \qquad Q = [k]P, \quad n \text{ prime}$$

Assume a retraction ρ : G → Gₙ with ρ(P)=P; set $F_j = \rho(B_j)$ for ambient factor-base elements B_j.

ELLIPTIC ENGINE

$$R_i = [a_i]P + [b_i]Q = \sum_j e_{ij} F_j$$

Constrained point decomposition

Summation polynomials and a lift/sign check

JACOBIAN ENGINE

$$[D_i] = [a_i]P + [b_i]Q = \sum_j e_{ij} F_j$$

Smoothness of reduced divisor classes

Factorization and partial-relation recombination

RELATION MATRIX

$$M\lambda = \mathbf{a} + k\mathbf{b} \pmod{n}, \qquad M \in \mathbb{F}_n^{r \times s}$$

LEFT-KERNEL EXTRACTION

$$\mathbf{w}^\top M = 0 \longrightarrow \mathbf{w}^\top \mathbf{a} + k\mathbf{w}^\top \mathbf{b} = 0 \longrightarrow k = -\frac{\mathbf{w}^\top \mathbf{a}}{\mathbf{w}^\top \mathbf{b}}$$

Required: $\mathbf{w}^\top \mathbf{b} \neq 0$ modulo n. Rank and denominator checks remain necessary after relation collection.

Example projector when #G=hn and gcd(h,n)=1: $\rho = [h^{-1} \bmod n] \circ [h]$. General primary decompositions require their own projector.

Divisor smoothness: factorization and support

An imaginary hyperelliptic model makes the infinity convention and the degree of a reduced divisor explicit.

$$C/\mathbb{F}_q:\ y^2 = f(x), \quad \deg f = 2g+1, \quad f \text{ squarefree}, \quad \operatorname{char}\mathbb{F}_q \neq 2$$

Use the unique point ∞ as base point; the divisor class group is J=Pic⁰(C).

REDUCED MUMFORD DATA

$$D = (u, v), \quad u \mid (f - v^2)$$

$$\deg v < \deg u \le g$$

u monic; take squarefree support for the displayed decomposition.

FACTOR u OVER $\mathbb{F}_q$

$$u = \prod_j u_j$$

$$d_j = \deg u_j$$

Each irreducible factor plus v mod u_j determines a place.

PRIME-DIVISOR CLASSES

$$F_j = [\mathfrak{p}_j - d_j\infty]$$

$$\deg \mathfrak{p}_j = d_j$$

Degree-one places correspond to rational points of C.

$$[D] = \sum_j [\mathfrak{p}_j - d_j\infty]\ ; \qquad D \text{ is smooth when all } F_j \in \mathcal{F}$$

A COUNTING HEURISTIC

$$\Pr[D\ \mathcal{F}\text{−smooth}] \approx \frac{1}{g!}\left(\frac{B}{q}\right)^g$$

B counts admissible signed degree-one classes; full B≈q gives ≈1/g!.

DOUBLE LARGE-PRIME VARIATION

Keep partial relations with ≤2 degree-one places outside the selected factor base. Eliminate those places by recombination.

Heuristic: uniform ambient classes, fixed g and distinct support. Restricted elliptic subfamilies need separate relation statistics.

Summation polynomials and Weil restriction

The difficult task is finding admissible solutions; polynomial identities alone do not establish a fast solver.

$$E/K:\ y^2 = x^3 + ax + b, \qquad K = \mathbb{F}_{q^d}, \quad \mathcal{F} = \{\pm P : x(P) \in \mathbb{F}_q\}$$

Assume characteristic >3 for the displayed short model. More general factor bases impose their own coordinate constraints.

POINT DECOMPOSITION

$$R = \sum_{i=1}^{m} \varepsilon_i P_i$$

$x_i \in \mathbb{F}q$; $\varepsilon_i \in \{+1, -1\}$

$$S_{m+1}(x_1, \ldots, x_m, x_R) = 0$$

$$\deg_{x_i} S_{m+1} = 2^{m-1} \quad (m \geq 2)$$

Roots encode geometric lifts; verify field membership and the signed group relation.

CHOOSE A BASIS 1, ω, …, ω^{d-1} OF K OVER $\mathbb{F}q$

$$S_{m+1} = \sum_{\nu=0}^{d-1} s_\nu(x_1, \ldots, x_m)\omega^\nu \longrightarrow s_0 = \cdots = s_{d-1} = 0$$

Dimension heuristic: m−d. At m=d the system is generically zero-dimensional only if these constraints are independent.

GENERIC EXTENSION-FIELD REGIME

$$\widetilde{O}(q^{2-2/d}), \qquad d \geq 3 \text{ fixed}$$

Gaudry’s heuristic regime: a suitable nondegenerate factor base and controlled decomposition cost.

SUBFIELD TARGET SPECIALIZATION

$$a, b, x_R, x_i \in \mathbb{F}_q \implies s_1 = \cdots = s_{d-1} \equiv 0$$

The canonical system collapses to one base-field equation.
A nonrational x-line changes this system and must be analyzed.

A summation-polynomial zero is not yet a verified relation in the intended group. Solver and independence assumptions are substantive.

Covers, norm maps and elliptic factors

A geometric cover gives a subgroup embedding on prime-to-degree torsion; it does not change the target order.

GENUS 2 · HYPERELLIPTIC

$$C_2 :\ Y^2 = X^6 + aX^2 + b$$

$f_2(X,Y)=(X^2,Y)$; $b\neq 0$ and $4a^3+27b^2\neq 0$

deg $f_2=2$ →

GENUS 1 · ELLIPTIC

$$E :\ y^2 = x^3 + ax + b$$

All maps and equations are over the same field $\mathbb{F}p$.

$$J_2 = \mathrm{Jac}(C_2) \sim E \times \widetilde{E}$$

$$\widetilde{E} :\ v^2 = bu^3 + au^2 + 1$$

The product decomposition is through a (2,2)-isogeny.

← f_2^*

f_{2*} →

$$\mathrm{Pic}^0(E) \cong E, \qquad Q = [k]P$$

$$f_2^* Q = [k] f_2^* P$$

If n is odd, the embedded subgroup still has order n.

$$f_{2*} \circ f_2^* = [2] \quad \Longrightarrow \quad \ker f_2^* \cap E[n] = \{O\} \ \ (n \text{ odd})$$

EXPLICIT FIBER DIVISOR

$$u(X) = X^2 - x_P, \qquad v(X) = y_P$$

$$f_2^*[P - O] = [f_2^{-1}(P) - (\infty_+ + \infty_-)]$$

These are fiber data on the even-degree model, not an arbitrary reduced class.

GENUS 3 · NON-HYPERELLIPTIC ALTERNATIVE

$$C_3 :\ W^4 = X^3 + aX + b, \quad f_3(X,W) = (X, W^2)$$

$$\mathrm{Jac}(C_3) \sim E \times \mathrm{Prym}(C_3/E)$$

Smooth plane quartic; dim Prym = 2. The degree-two norm identity also holds.

Characteristic >3. ECCp-131 satisfies the smoothness conditions. The symbol ~ denotes an isogeny, not equality of groups or polarizations.

Frobenius decomposes the extension-field groups

For ECCp-131, the target prime factor is confined to the base-field summand in both extensions.

$$\pi = \mathrm{Frob}_p, \quad t = p + 1 - n, \quad |E(\mathbb{F}_p)| = n \text{ prime}$$

For the published curve: gcd(n,n′)=gcd(n,M_3)=1, n≠2,3. These are checked arithmetic conditions.

QUADRATIC EXTENSION

$$N_2 = nn', \qquad n' = p + 1 + t$$

$$e_+ = \frac{1+\pi}{2}, \qquad e_- = \frac{1-\pi}{2}$$

$$E(\mathbb{F}_{p^2}) \cong E(\mathbb{F}_p) \times E'(\mathbb{F}_p)$$

Here the group has odd order, so division by 2 is defined.

$$e_+ E'(\mathbb{F}_p) = \{O\}$$

Set $\rho_2 = e_+$. For $R \in E(\mathbb{F}p)$, the twist component $e_- R$ is zero.

CUBIC EXTENSION

$$N_3 = nM_3, \quad M_3 = p^2 - p + 1 + (p+1)t + t^2$$

$$T_3 = 1 + \pi + \pi^2, \qquad \rho_3 = [3^{-1} \pmod n] T_3$$

$$E(\mathbb{F}_{p^3}) \cong E(\mathbb{F}_p) \times \ker T_3$$

$$\{P \in E(\mathbb{F}_{p^3}) : x(P) \in \mathbb{F}_p\} = E(\mathbb{F}_p) \setminus \{O\}$$

For a finite point with x∈𝔽p, the y-coordinate has degree dividing both 2 and 3, hence lies in 𝔽p.

NONRATIONAL FACTOR BASES CAN ESCAPE THESE CANONICAL BASES

$$R = \sum_j e_j B_j \quad \Longrightarrow \quad R = \sum_j e_j \rho_d(B_j), \quad R \in E(\mathbb{F}_p),\ d = 2, 3$$

Projection constrains any resulting relation; it does not rule out every algebraic improvement in how relations are generated.

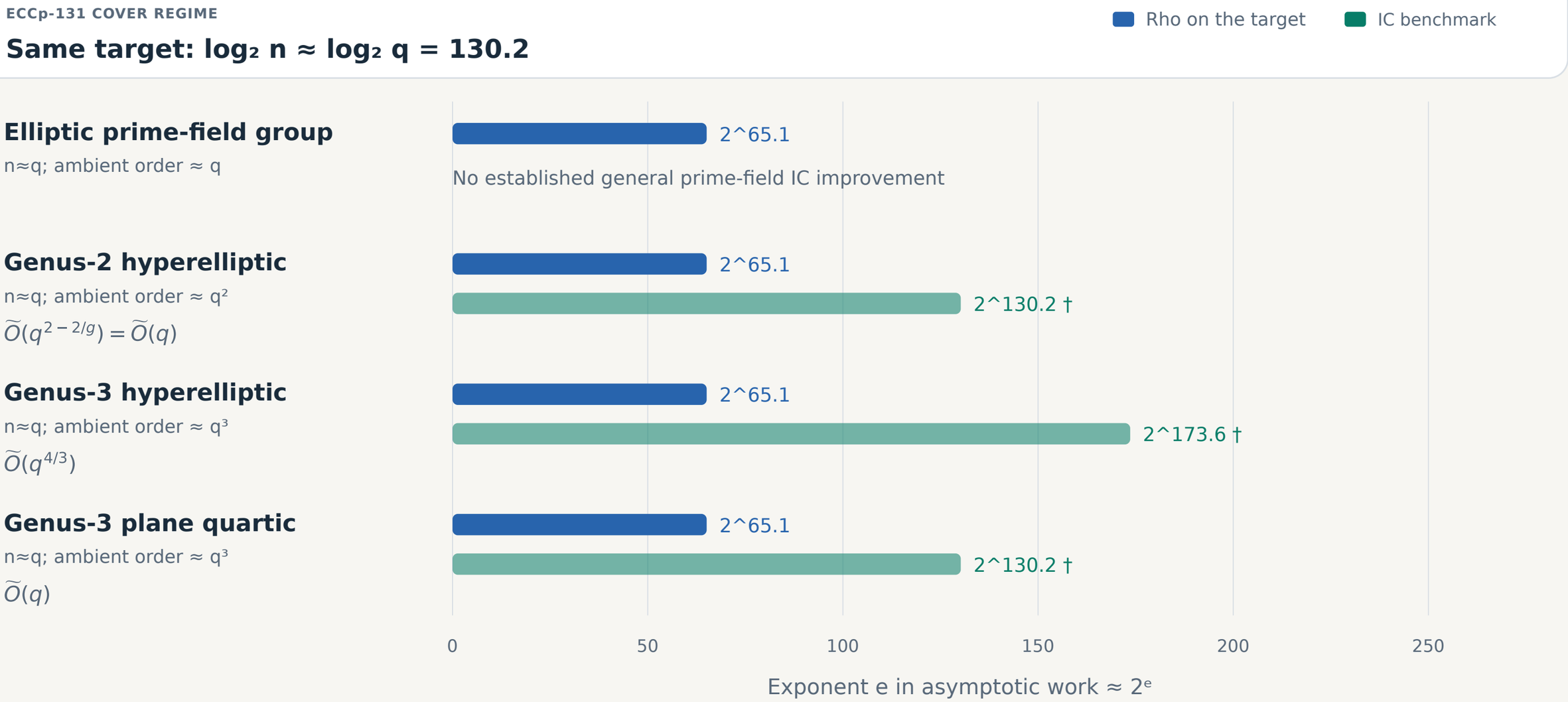
The displayed direct products use the checked ECCp-131 hypotheses. Without them, intersections and primary-component issues require care.

Complexity regimes: ambient group or target subgroup?

Compare asymptotic work exponents with the prime subgroup order n stated explicitly.

ECCp-131 COVER REGIME

Same target: $\log_2 n \approx \log_2 q = 130.2$



† Ambient IC benchmark: split-Jacobian hypotheses and restricted-subgroup relation statistics must be rechecked.

The target subgroup retains n≈q under the covers. Its rho exponent is therefore $\log_2(n)/2$ in every row.

Fixed g; constants and logarithmic factors suppressed. Algorithmic hypotheses and operation costs differ; these are not runtime predictions.

Worked extraction: three rows can determine k—or fail

The relations below are constructed and checked in an actual toy elliptic group; they do not measure relation-search cost.

$E/\mathbb{F}_{571} : y^2 = x^3 + 508x + 137, \quad |E| = 557, \quad P = (517, 168)$

Input Q=(560, 245). The factor base has two unknown logs. All three rows are valid group identities.

$M\lambda = \mathbf{a} + k\mathbf{b} \pmod{557}$

M

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

a

$$\begin{bmatrix} 494 \\ 469 \\ 332 \end{bmatrix}$$

b

$$\begin{bmatrix} 2 \\ 3 \\ 7 \end{bmatrix}$$

rank M = 2

rank [M | −b] = 3

3 rows in both modes

$\mathbf{w}^\top = (-1, -1, 1), \quad \mathbf{w}^\top M = 0$

$\mathbf{w}^\top\mathbf{a}$ = 483 **$\mathbf{w}^\top\mathbf{b}$ = 2**

$$k = -\frac{\mathbf{w}^\top \mathbf{a}}{\mathbf{w}^\top \mathbf{b}}$$

k = −483 · 2⁻¹ = 37 (mod 557)

A nonzero target-column pairing turns a factor-base dependency into the logarithm.

Toggle the target column in the viewer. Keeping three valid rows does not guarantee that their target coefficients carry information.

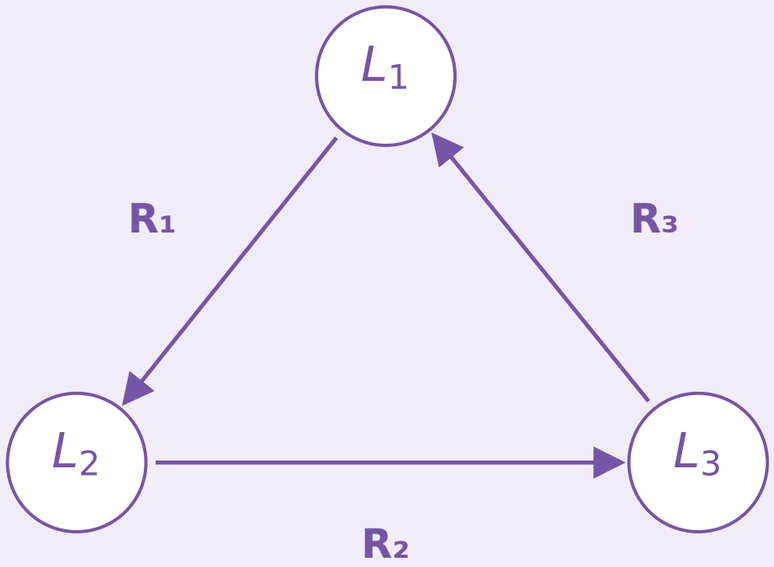
Partial relations: cancel the large-prime columns

An oriented graph is a bookkeeping device for signed divisor-class relations.

$$R_i = [a_i]P + [b_i]Q = \sum_j e_{ij} F_j + \sum_\ell A_{i\ell} L_\ell$$

F_j lie in the selected factor base. L_ℓ are exceptional degree-one classes outside it; retain their signs.

A SIGNED LARGE-PRIME TRIANGLE



THE EDGE EQUATIONS

$$R_1 = F_1 + L_1 - L_2$$

$$R_2 = F_2 + L_2 - L_3$$

$$R_3 = F_3 + L_3 - L_1$$

$$\mathbf{z}^\top A = 0, \quad \mathbf{z} = (1, 1, 1)$$

Add the rows; every exceptional class cancels.

$$R_1 + R_2 + R_3 = F_1 + F_2 + F_3$$

$$[363]P + [7]Q = F_1 + F_2 + F_3 \quad \text{(toy order 557)}$$

A cycle in an unsigned graph is not, by itself, a verified relation. Use the full signed coefficient matrix when eliminating exceptional classes.

Fiber smoothness is a structured distribution

A rational divisor class can be supported on rational points or on a nonrational closed point.

$$f : C_2 \longrightarrow E, \quad (X, Y) \longmapsto (X^2, Y), \qquad u_P(X) = X^2 - x_P$$

The published ECCp-131 points have χ(xP)=+1 and χ(xQ)=−1. Their pullbacks are both $\mathbb{F}_p$-rational classes.

ACTUAL ECCp-131 FIBERS

$$B_{\pm} = (\pm r, y_P), \qquad r^2 = x_P$$

$f^{-1}(P)=\{B_+,B_-\}$: two rational points; uP splits over $\mathbb{F}_p$.

$$f^{-1}(Q) = \{A_+, A_-\}, \quad \pi(A_+) = A_-$$

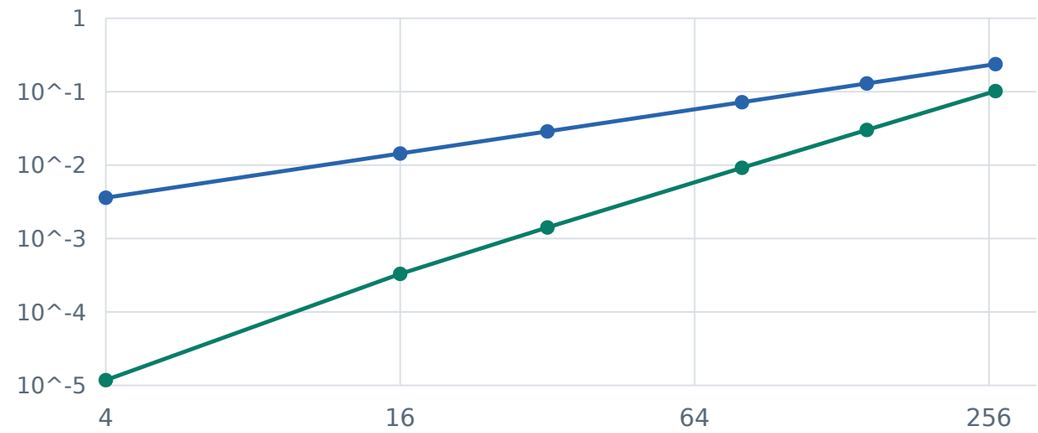
One degree-two place: uQ is irreducible over $\mathbb{F}_p$.

$$[A_+ + A_- - (\infty_+ + \infty_-)] \in J(\mathbb{F}_p)$$

Rationality of the class does not require rational support.

EXACT TOY CENSUS: p=571, n=557

$|J(\mathbb{F}_p)| = 557 \cdot 609 = 339213$; both axes logarithmic.



$\Pr_E = \frac{B}{2(n-1)}$ $\qquad$ $\Pr_J = \frac{B^2/2 - B}{|J(\mathbb{F}_p)|}$

$$\rho_C = [2^{-1} \pmod n] f_*, \qquad \rho_C f^* = \mathrm{id}$$

$$\rho_C[B_+ - \infty_+] + \rho_C[B_- - \infty_-] = [2^{-1}]P + [2^{-1}]P = P$$

Fiber splitting gives correlated projected summands. It does not reveal their factor-base logarithms.

Census scope: F_B is closed under X↦−X and Y↦−Y, excludes branch/infinity points; S_B uses two distinct X-coordinates. See notes for counts.

From more roots to less total work

A change in relation yield must be evaluated together with verification, recombination, rank and matrix costs.

$$T = T_{\text{setup}} + T_{\text{collect}} + T_{\text{filter}} + T_{\text{LA}} + T_{\text{indiv}}$$

Use measured operation costs on the same target subgroup and factor-base setup before comparing methods.

A PHASE-AVERAGE COLLECTION MODEL

$$c_{\text{eff}} = \frac{c_{\text{trial}}}{p_{\text{valid}}\eta}, \qquad T_{\text{collect}} \approx R_{\text{req}} c_{\text{eff}}$$

pvalid = probability of a verified relation per trial.
η = effective information yield per verified relation.
Assume pvalid,η>0; rates may change during collection.

FILTERED SPARSE LINEAR ALGEBRA

$$T_{\text{LA}} \sim s\,C_{\text{mv}}, \qquad C_{\text{mv}} \sim r\overline{w}\,c_n$$

$$\text{memory} \sim O(r\overline{w} + s) \text{ field entries}$$

Schematic Krylov scale after filtering to r≈s.
c_n is modular-arithmetic cost; $\bar{w}$ is average row weight.

ILLUSTRATIVE MODEL · NOT CHALLENGE ATTACK MEASUREMENTS

Collection cost per effective row, in normalized work units



Alternative B: 2.00× the baseline collection cost per effective row.

The collection formula is a diagnostic model, not a theorem. Matrix filtering can change dimensions and row weight; include every phase in total work.